Group Participation Sheet

Prof. Mouhamed Abdulla, School of Applied Computing, FAST
Sheridan College Institute of Technology and Advanced Learning, GTA, ON, Canada

A portion of this course is devoted to group work. This **Group Participation Sheet** (GPS), hereinafter referred to as the **GPS Sheet**, will specifically indicate the group members that actively contributed to the assignment, lab project, deliverable, or report. All names indicated on this GPS sheet will receive the same grade for the attached group work. Group members with names and signatures not shown on this GPS sheet will receive a grade of "**zero**" for the mentioned assessment. Any concerns regarding this matter should be brought to the professor's attention by the team leader prior to the submission deadline.

Please fill the table below electronically. Group members that participated in this evaluation should **write their names** and append their **signature**. The signed GPS sheet should be added as the **first page of all group evaluations**. This should be **applied** to all evaluations uploaded on SLATE as **softcopy**, and printed/stapled and presented as **hardcopy** to your professor.

Clear Form Print Form

Course Title:	PROG-50139 - Python Programming for AI	
Evaluation Name:	PROG-50139 - A2 (10%, Group Work)	
Team Number:	N/A	
Referencing and Citation Style:	*APA Style* ○ *IEEE Style* ○ *N/A* ●	
Group Members:	*First Name and Family Name of group members:*	*Signature of group members:*
	(1) Sweta Chowdary (team leader)	Sweta chowdary
	(2) Maheshwar Tandon	Maheshwar Tandon
	(3) Anisha Chaudhary	
	(4) Oluwadolapo Phillips	
	(5) Vasu Vasu	Vasu Vasu
Submission Date:	03-31-2026	

Python Programming for AI
Assignment 2

a) Generate N = 25 random uniform integers from the set $S = [-7, 3) = \{n \in \mathbb{Z} \mid -7 \leq n < 3\}$, and store these values in a 1D horizontal tensor $\mathbf{H1}$. Hint: You may find the function np.random.randint from the NumPy library useful

H1 is a horizontal 1D tensor (matrix) containing N random integers selected from the set
S = { n ∈ ℤ | S_start ≤ n < S_end }.

So, H1 can be written as:

H1 = [h_1, h_2, h_3, ..., h_N], where each $h_i \in S$

This means H1 is a row matrix with N elements.

(b) Output the shape and size of $\mathbf{H1}$. Note: Think of $\mathbf{H1}$ as a matrix (not a vector).

Since H1 is a horizontal matrix:

Shape of H1 = (1, N)
Size of H1 = N

(c) Create a new horizontal 1D tensor $\mathbf{H2}$ that store the entries of $\mathbf{H1}$ in the reverse order (i.e., [1, 2, 3] becomes [3, 2, 1]). Find the shape and size of $\mathbf{H2}$. Note: Again, think of $\mathbf{H2}$ as a matrix (not a vector).

H2 is formed by reversing the order of elements in H1.

H2 = [h_N, h_(N-1), ..., h_1]

Even though the order changes, the number of elements remains the same.

Shape of H2 = (1, N)
Size of H2 = N

(d) Change the 1D horizontal tensors $\mathbf{H1}$ and $\mathbf{H2}$ into 1D vertical tensors $\mathbf{V1}$ and $\mathbf{V2}$. Find the shape and size
of $\mathbf{V1}$ and $\mathbf{V2}$.

V1 and V2 are obtained by transposing H1 and H2.

V1 = transpose of H1 → (N,1)
V2 = transpose of H2 → (N,1)

So:

Shape of V1 = (N, 1), Size = N
Shape of V2 = (N, 1), Size = N

(e) Assuming we think of H1 and H2 as vectors, then apply their dot product using matrices V1 and V2 by way of matrix multiplication. Store the result in D and find its shape and size.

To compute the dot product using matrices, we use:

$D = V1^T V2$

Since V1 has shape (N,1), its transpose $V1^T$ has shape (1,N).
V2 has shape (N,1), so:

$(1\times N)(N\times 1) = (1\times 1)$

So, D is a single value (scalar).

In expanded form:

D = Sum from i = 1 to N of (h_i × h_(N - i + 1))

This happens because H2 is the reverse of H1, so corresponding elements are paired in reverse order.

Shape of D = (1,1)
Size of D = 1

(f) Assuming we think of H1 and H2 as vectors, then, find the cosine similarity C. Hint: You may find the function np.linalg.norm from the NumPy library useful.

Cosine Similarity C

Cosine similarity measures how similar two vectors are by comparing their direction.

$C = D / (\|V1\| \times \|V2\|)$

Expanded:

$C = [\text{Sum from } i = 1 \text{ to } N \text{ of } (h_i \times h_{(N - i + 1)})]$
divided by
$[\text{sqrt}(\text{Sum from } i = 1 \text{ to } N \text{ of } h_i^2) \times \text{sqrt}(\text{Sum from } i = 1 \text{ to } N \text{ of } h_{(N - i + 1)}^2)]$

Here:

- The numerator is the dot product
- The denominator normalizes by the lengths of the vectors

(g) Do the vectors H1 and H2 have low, neutral, or high similarity? Explain your logic.

Similarity Interpretation

The similarity between H1 and H2 is low to moderate.

Even though both vectors contain the same values, H2 is the reverse of H1. This means the elements are not aligned in the same positions. Since cosine similarity depends on how well corresponding elements match, this misalignment reduces the similarity.

(h) Matrix Multiplications and Observations

We compute the following matrix multiplications:

1. $V1\ V2^T$

V1 has shape (N,1) and $V2^T$ has shape (1,N)

So:
$(N\times1)(1\times N) = (N\times N)$

This results in an $N \times N$ matrix.

Each element is computed as:
Entry $(i, j) = V1_i \times V2_j$

This means every element of V1 is multiplied with every element of V2, forming an outer product.

2. $V2^T\ V1$

$V2^T$ has shape (1,N) and V1 has shape (N,1)

So:
$(1\times N)(N\times 1) = (1\times 1)$

This results in a single value (scalar), which is the dot product.

In expanded form:

$V2^T V1$ = Sum from i = 1 to N of ($h_i \times h_{(N - i + 1)}$)

3. $V2\ V1^T$

V2 has shape (N,1) and $V1^T$ has shape (1,N)

So:
$(N\times 1)(1\times N) = (N\times N)$

This also results in an $N \times N$ matrix.

Each element is:
Entry $(i, j) = V2_i \times V1_j$

🔍 Observation

From these results, we observe that:

- $V2^T V1$ gives a scalar (dot product)
- $V1\ V2^T$ and $V2\ V1^T$ give matrices (outer products)
- The matrices $V1\ V2^T$ and $V2\ V1^T$ are not the same

This shows that matrix multiplication is not commutative, meaning:

$V1\ V2^T \neq V2\ V1^T$

(i) Matrix Construction from N Values

Let N values be generated from the set S.

Assume N is a perfect square, so we can form a matrix of size $n \times n$, where:

$n = \sqrt{N}$

Matrix 1 (Main Diagonal)

A matrix where the given values are placed along the main diagonal.

Entry (i, i) = value_i
All other entries = 0

Matrix 2 (Secondary Diagonal)

A matrix where values are placed along the other diagonal.

Entry (i, n − 1 − i) = value_i
All other entries = 0

Matrix 3 (Sum of Matrix 1 and Matrix 2)

This matrix is obtained by adding Matrix 1 and Matrix 2.

Matrix 3 = Matrix 1 + Matrix 2

If both diagonals overlap (in case of odd n), the center value is added.

Matrix 4 (Serpent Pattern)

This matrix is filled column-wise in a zig-zag (serpent) pattern.

- Even-indexed columns are filled from top to bottom
- Odd-indexed columns are filled from bottom to top

This creates an alternating vertical pattern across columns